Does a Pending-Sector Warning Predict Hard-Drive Failure?

A survival analysis of Backblaze Drive Stats, Q1 2026

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Question and hypotheses

- Drives report SMART health counters; SMART 197 = Current Pending Sector Count (unreadable sectors awaiting reallocation)
- Question: is a nonzero SMART 197 on 1 Jan 2026 associated with a later recorded failure — after accounting for age?
- H_0: baseline SMART 197 status has no association with the subsequent failure hazard
- H_a: baseline SMART 197 status is associated with the subsequent failure hazard
- Survival framing: follow-up lengths differ, most drives never fail (censoring), some leave observation early

Data: Backblaze Drive Stats, Q1 2026

- Public daily snapshots of every drive: date, serial, model, failure flag (0/1), ~90 SMART pairs
- Q1 2026: 90 daily files, 1 Jan – 31 Mar, 30.6 million drive-days (archive hash and all file checksums verified)
- One drive model only: Seagate ST12000NM0008 (12 TB) — 18,650 drives, 129 failure flags in Q1
  - removes manufacturer / firmware / capacity differences from the comparison
- Outcome = Backblaze's recorded failure flag (definition matters — slide 10)

Cohort, outcome and censoring

Baseline SMART 197	Drives	Failures	Censored 31 Mar	Exits
= 0	18,371	83	18,195	93
> 0	162	28	131	3
Total	**18,533**	**111**	**18,326**	**96**

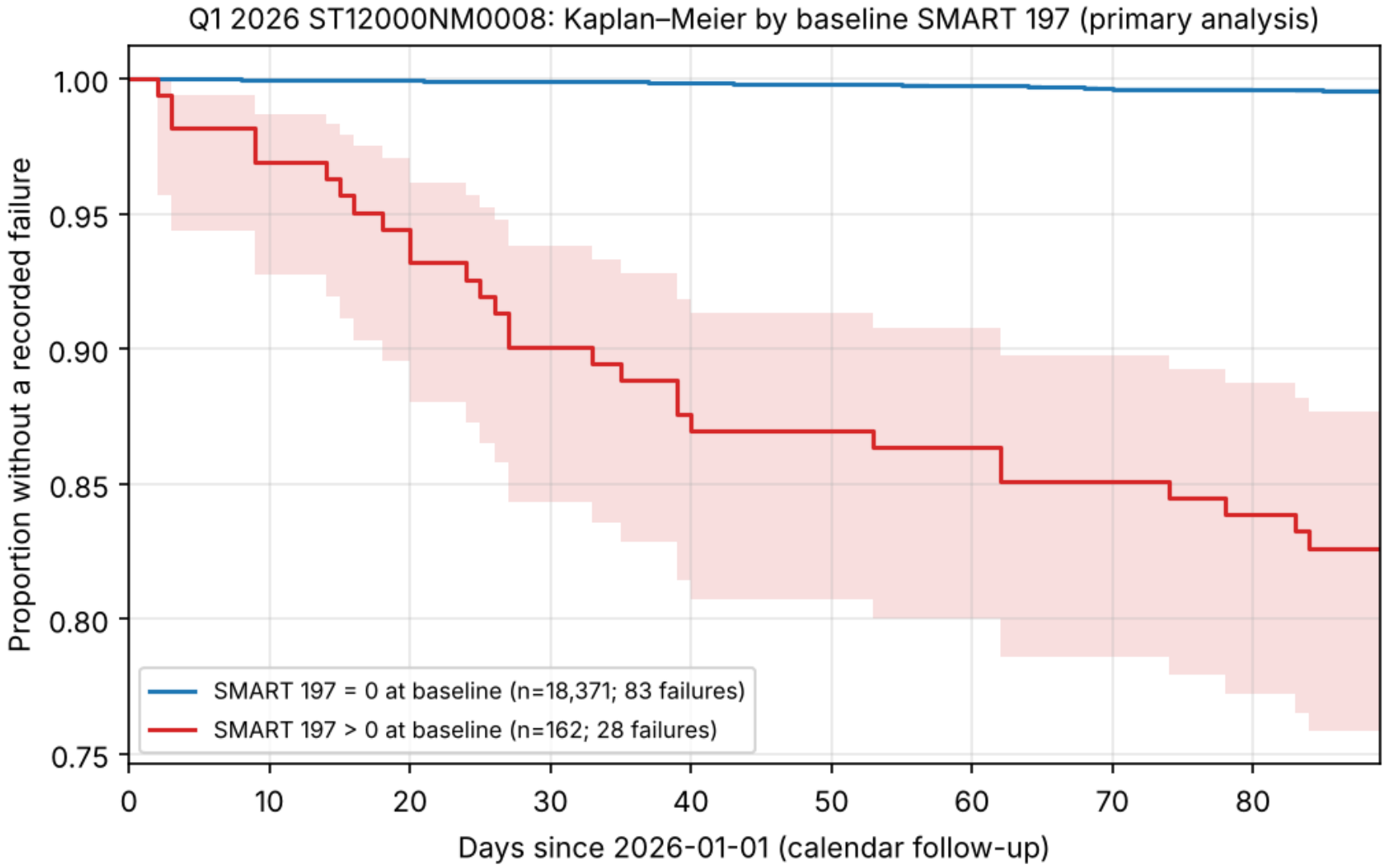
- Day 0 = 1 Jan 2026; follow-up to day 89
- Excluded: 116 later arrivals, 1 failure on day 0
- Event: first failure flag after day 0
- Unexplained exits are censored, never failures
- Exposure from the 1 Jan row only

- Data checks: no duplicate drive-days, no rows after a failure, no missing baseline SMART 197 / 5 / 9
- 10.2% of drives have an observation gap (22% in the SMART 197 > 0 group)

Methods

- Kaplan–Meier by baseline group; 95% CI by Greenwood, log(−log) scale
- Cox PH: $h(t|x) = h_0(t) \cdot \exp(\beta x)$, Efron ties; Wald test of $H_0: \beta = 0$
  - PH check: scaled-Schoenfeld test (Grambsch–Therneau) — reported, not used to change the model
- Sensitivity S1: 93 exits that reappeared in April → censored at day 89
- Sensitivity S2: remove gap days from the risk set (start–stop intervals)
- Age adjustment: + baseline powered years (SMART 9 ÷ 8,766), since install dates are unknown
- Rules and plan committed before looking at failures by group; lifelines results cross-checked in statsmodels

Kaplan–Meier: drives with a pending-sector warning fail far more often



How big is the difference?

Proportion without a recorded failure	Day 30	Day 60	Day 89
SMART 197 = 0 (n = 18,371)	0.9989	0.9976	0.9955 (0.9944–0.9964)
SMART 197 > 0 (n = 162)	0.9007	0.8635	0.8262 (0.7583–0.8766)

- ≈ 17 in 100 warned drives vs < 1 in 200 others had a recorded failure within 3 months
- Warned drives: < 1% of the fleet, but 25% of the failures (28 of 111)

Cox model: reject H_0

Analysis	HR (SMART 197 > 0 vs 0)	95% CI	Wald p
Primary	**43.2**	**28.1 – 66.3**	**1.7×10^{-66}**
S1: exits reclassified	43.2	28.1 – 66.3	1.7×10^{-66}
S2: gap days removed	44.1	28.7 – 67.7	3.5×10^{-67}
Adjusted for powered years	43.6	28.4 – 66.9	9.5×10^{-67}

- Powered years: HR 1.11 per year (0.85–1.43), p = 0.44 — but almost all drives are 5.5–6.2 years
- Decision: reject H_0 — with and without age adjustment

The risk is front-loaded: proportional hazards fail

- PH test (scaled Schoenfeld): $\chi^2 = 7.29$, $p = 0.0069$
- Warned drives: ~10% fail in the first 30 days, then ~4% per 30 days
- HR = 43 is a quarter-average of a ratio that is larger early and smaller later
- We report KM curves and landmark survival alongside the HR; model left as pre-specified

What the result does — and does not — mean

- Backblaze counts proactive replacements (triggered by SMART stats, FSCK checks …) as failures
  - Q3 2024: 721 of 1,361 failures (53%) were proactive; the public 0/1 flag does not distinguish them
- → part of the association is built into the outcome: strong association, not a causal effect
- Consistent with Pinheiro et al. (Google, 2007): after a first pending-sector count, 16× more likely to fail within 60 days
  - …but > 56% of their failed drives showed no strong SMART signal — not a stand-alone predictor

Limitations

- Outcome includes SMART-informed proactive replacements (circularity)
- Prevalent cohort, no install dates; age proxy (powered hours) spans a narrow range
- Daily snapshots: 21 of 44 post-gap failures flagged on the return day
- 4 'exited' drives were flagged as failed in early April — exits not fully uninformative
- 110 of 162 warned drives also had reallocated sectors (SMART 5) — not specific to pending sectors
- One model, one quarter, 28 exposed failures → wide confidence interval

Conclusion

- A nonzero SMART 197 on 1 Jan marks < 1% of drives with a far higher 3-month recorded-failure rate
- HR 43.2 (28.1–66.3); 43.6 after adjusting for powered-on age — H_0 rejected
- Robust to exit and gap handling; excess risk concentrated in the first month
- Interpretation: strong association with recorded removal-or-failure — not causal, not a validated predictor

Thank you — questions?

Backup: safeguards and reproducibility

- Commit order: decision rules (e3cf5dd) → group counts (4259dc3) → analysis plan (2b91f38) → results (28cb715)
- Age model: plan (e117f03) committed before fit (714fb97)
- Cohort rebuilt from a re-downloaded, SHA1-verified archive: byte-identical
- lifelines vs statsmodels: coefficients agree to $< 10^{-6}$; 49 automated tests pass
- Repository: aayush-0131/backblaze-survival-analysis (no raw or drive-level data committed)